
ADDENDUM TO PDSTRIP DOCUMENTATION

FOREWORD

The PDSTRIP program has been modified so as to obtain as output the parameters necessary for running the SIMBAD program, namely for a certain number of frequencies and wave directions: added masses, radiation dampings, wave and drift forces. This addendum is not intended to replace the documentation of the PDSTRIP program but to supplement it with some additional information. Please refer to the PDSTRIP documentation for any further information.

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THE INPUT DATA ARE READ FROM THE FILE “PDSTRIP.INP”.

ANOTHER FILE CONTAINS THE SECTION OFFSETS OF THE SHIP. THE NAME IS DEFINED IN THE PREVIOUS FILE, AS FOR EXAMPLE “GEOMET.OUT”.

RESULTS ARE WRITTEN INTO THE OUTPUT FILE “PDSTRIP.OUT” AND IN ANOTHER DATA FILES, BUT ONLY THE “PDSTRIP.OUT” FILE IS FURTHER USED BY THE **SIMBAD** PRE-PROCESSING PROGRAM.

THE INPUT IS SPECIFIED IN A COORDINATE SYSTEM WHERE THE X AXIS IS POINTING FORWARD, Y TO PORT AND Z UPWARD. IN THE **PDSTRIP** DOCUMENTATION, IT IS SPECIFIED THAT THE COORDINATE ORIGIN IS, UNLESS SPECIFIED OTHERWISE, LOCATED AT THE INTERSECTION OF MIDSHIP PLANE, MIDSHIP SECTION AND BASE-LINE (KEEL). IT IS NEVERTHELESS RECOMMENDED TO LOCATE THE X AND Y COORDINATES OF THE ORIGIN AT THE POSITION OF THE CENTRE OF GRAVITY ($X=Y=0$), SO AS TO POSITION THE X COORDINATES OF THE SECTIONS ACCORDING TO COG.

IN THE OUTPUT FILE “PDSTRIP.OUT”, THE PARAMETERS NECESSARY FOR RUNNING THE **SIMBAD** PROGRAM (ADDED MASSES, RADIATION DAMPINGS, WAVE AND DRIFT FORCES) ARE GIVEN IN THE SAME TYPE OF COORDINATE SYSTEM AS FOR THE INPUT BUT WITH THE ORIGIN LOCATED AT THE COG(*).

(*)AND NOT IN THE COORDINATE SYSTEM 1, 2, 3 AS SPECIFIED IN THE **PDSTRIP** DOCUMENTATION

INFORMATION FOR THE FILE “PDSTRIP.INP”:

AN EXAMPLE OF FILE IS GIVEN HEREAFTER. ONLY FEW PARAMETERS NEED TO BE CHANGED BETWEEN TWO CALCULATIONS FOR TWO TYPES OF VESSELS (APART FROM THE SECTION OFFSET FILE).

```
-20 t t t
PC TEU 3000
9.81 1.025 8.3 -8.3 999.
7 90. 60. 30. 0. -30. -60. -90.
geomet.out

f
39107.0 0.0 0.0 12.0 124.6 3186.6 3186.6 0.0 0.0 0.0
0 0 0 0 0
0 0 0 0 0
0 0 0 0 0
0 0 0 0 0
0 0 0 0 0
0 0 flow separation
0.03 3.0 wave steepness; max wave height
0.8 0.6 0.8 0.6 0.8 0.6 0.8 0.6 0.8 0.6
0.8 0.6 0.8 0.6 0.8 0.6 0.8 0.6 0.8 0.6
0.8 0.6 0.8 0.6 0.8 0.6 0.8 0.6 0.8 0.6
0.8 0.6 0.8 0.6 0.8 0.6 0.8 0.6 0.8 0.6
0.8 0.6 0.8 0.6 0.8 0.6 0.8 0.6 0.8 0.6
0.8 0.6 0.8 0.6 Cy Cz
0 fin
0 sails
0 forces depending on motions
0.0 0.0 0.0 0.0 0.0 suspended weight
0 motion points
```

TITLE FOR THE TYPE OF SHIP

DRAUGHT AND DEPTH FROM THE BASE-LINE (KEEL)

WAVE ENCOUNTER ANGLES. ORDER TO BE KEPT LIKE THIS FOR THE PRE-PROCESSING PROGRAM (BETWEEN +90 AND -90).

MASS OF SHIP. COORDINATES OF THE CENTRE OF GRAVITY. INERTIAL RADII.

NUMBER OF COEFFICIENTS TO BE INCREASED OR DECREASED AS A FUNCTION OF NUMBER OF OFFSET SECTIONS

INFORMATION FOR THE FILE “PDSTRIP.INP” (FOLLOWING):

```
99
1000.0      750.0248    514.0973    389.3734    312.2181    259.3743
 220.8418    191.5603    168.2591    149.1822    133.3587
 120.0826    108.4242     98.3848     89.4751     81.5469
  74.4736     68.1476     62.4175     57.2253     52.5182
  48.3288     44.5147     41.0719     38.0135     35.2345
  32.7270     30.4780     28.4349     26.5906     24.9201
  23.3887     22.0065     20.7434     19.5756     18.5132
  17.5263     16.6243     15.7827     15.0104     14.2934
  13.6206     12.9997     12.4152     11.8741     11.3630
  10.8886     10.4392     10.0208      9.6271      9.2528
   8.9031      8.5699      8.2578      7.9598      7.6802
   7.4126      7.1611      6.9222      6.6930      6.4769
   6.2692      6.0732      5.8845      5.7061      5.5357
   5.3714      5.2156      5.0652      4.9225      4.7845
   4.6534      4.5265      4.4058      4.2899      4.1774
   4.0703      3.9664      3.8672      3.7709      3.6790
   3.5895      3.5041      3.4217      3.3415      3.2647
   3.1899      3.1182      3.0483      2.9814      2.9160
   2.8534      2.7944      2.7340      2.6755      2.6189
   2.5690      2.5157      2.4640 wave length
```

NUMBER OF WAVE LENGTHS (FREQUENCIES) FOR WHICH THE PARAMETERS SHOULD BE DETERMINED (ABOUT 100 AT LEAST) . THE FREQUENCY RANGE SHOULD BE IN BETWEEN 0.01 AND 5 RAD/s. ACCORDING TO THE PDSTRIP CALCULATION RESULTS (SEE HEREAFTER) , IT IS SOMETIMES NECESSARY TO INCREASE THE LOWER LIMIT OF FREQUENCY RANGE (UP TO 0.05 TO 0.1 RAD/s) AND/OR DECREASE THE UPPER LIMIT OF FREQUENCY RANGE (DOWN TO 4.8 TO 4.9 RAD/s)
(WAVE LENGTHS ALWAYS IN DECREASING ORDER OR FREQUENCIES IN INCREASING ORDER)

```
1
0.0 t      1 speed

2
0 1
100 1

2
6
90
3
3.3

0/
```

INFORMATION FOR THE OFFSET SECTION FILE:

27	F	8.300000		
-112.7035		5	0	
-5.205000	-1.731000		0.000000E+00	1.731000
8.300000	7.970000		7.812000	7.970000
-107.7035		17	0	
-10.65500	-8.729000		-7.995000	-5.707000
-2.988000	-1.400000		-0.2920000	0.000000E+00
1.400000	2.988000		5.331000	5.707000
8.729000	10.65500			7.995000
8.300000	8.018000		7.922000	7.667000
7.407000	7.262000		7.163000	7.137000
7.262000	7.407000		7.631000	7.667000
8.018000	8.300000			7.922000
-97.70351		21	0	
-14.37800	-13.26200		-11.73100	-10.57100
-8.570000	-7.247000		-5.583000	-4.059000
0.000000E+00	2.425000		4.059000	5.583000
8.570000	9.747000		10.57100	11.73100
14.37800				13.26200
8.300000	7.921000		7.531000	7.281000
6.890000	6.659000		6.400000	6.185000
5.688000	5.973000		6.185000	6.400000
6.890000	7.109000		7.281000	7.531000
8.300000				7.921000
-87.70351		29	0	
-15.50400	-15.12800		-14.59500	-13.82000
-11.23300	-9.291000		-8.168000	-7.040000
-3.352000	-1.960000		-1.340000	-0.7310000
0.000000E+00				
0.7310000	1.340000		1.960000	3.352000
7.040000	8.168000		9.291000	11.23300
13.82000	14.59500		15.12800	15.50400
8.300000	7.920000		7.530000	7.106000
6.179000	5.690000		5.440000	5.204000
4.500000	4.258000		4.172000	4.119000
4.119000	4.172000		4.258000	4.500000
5.204000	5.440000		5.690000	6.179000
7.106000	7.530000		7.920000	8.300000
-77.70351		37	0	
-15.85200	-15.71700		-15.51400	-15.18200
-13.88200	-12.81900		-11.55300	-10.18800
-7.553000	-6.351000		-5.191000	-3.745000
-2.017000	-1.277000		-0.6560000	0.000000E+00
1.277000	2.017000		2.843000	3.745000
6.351000	7.553000		8.829000	10.18800
12.81900	13.88200		14.65900	15.18200
15.71700	15.85200			15.51400
8.300000	7.920000		7.530000	7.106000
6.179000	5.690000		5.204000	4.741000
3.964000	3.653000		3.375000	3.046000
2.683000	2.545000		2.443000	2.442000
2.545000	2.683000		2.853000	3.046000
3.653000	3.964000		4.323000	4.741000
5.690000	6.179000		6.654000	7.106000
7.920000	8.300000			7.530000
-67.70351		45	0	
-15.98400	-15.94000		-15.87200	-15.74700
-15.18500	-14.68900		-14.07000	-13.36800

USE ALWAYS "SYM = F (FALSE)", FOR WHICH THE SECTIONS ARE DESCRIBED BY OFFSET POINTS BEGINNING AT THE WATERLINE ON STARBOARD AND ENDING AT THE WATERLINE ON PORT SIDE. THE WAVE DRIFT FORCES CAN BE DETERMINED ONLY IF "SYM = FALSE".

INFORMATION FOR THE OFFSET SECTION FILE:

```
1.491000    1.860000    2.213000    2.543000    2.812000
2.963000    2.984000    2.921000    2.837000    2.701000
2.493000    2.193000    1.790000    1.331000    0.8780000
0.5030000   0.2620000   0.1420000   0.1070000   0.1350000
8.300000    7.530000    7.106000    6.654000    6.179000
5.690000    5.208000    4.741000    4.323000    3.964000
3.653000    3.375000    3.113000    2.594000    2.335000
1.817000    1.333000    0.9450000   0.6640000   0.4560000
0.2930000   0.1640000   0.0000000E+00 0.1640000   0.2930000
0.4560000   0.6640000   0.9450000   1.333000    1.817000
2.335000    2.594000    3.113000    3.375000    3.653000
3.964000    4.323000    4.741000    5.208000    5.690000
6.179000    6.654000    7.106000    7.530000    8.300000
119.7965    41    2    3    38
-1    -0.5    0.00E+00    0    -0.698
-1.268    -1.543    -2.096    -2.495    -2.739
-2.891    -2.982    -3.051    -2.984    -2.898
-2.645    -2.308    -1.876    -1.279    -0.525
0.00E+00    0.525    1.279    1.876    2.308
2.645    2.898    2.984    3.051    2.982
2.891    2.739    2.495    2.096    1.543
1.268    0.698    0    0.00E+00    0.5
1
8.3    8.1    7.9    4.943    4.902
4.763    4.651    4.323    3.964    3.653
3.375    3.113    2.594    2.075    1.817
1.333    0.945    0.664    0.466    0.37
0.353    0.37    0.466    0.664    0.945
1.333    1.817    2.075    2.594    3.113
3.375    3.653    3.964    4.323    4.651
4.763    4.902    4.943    7.9    8.1
8.3
```

FOR A BULB, IT IS NECESSARY TO INTRODUCE AN UNCONNECTED PART (SEE PDSTRIP DOCUMENTATION P. 8)

RESULTS ARE WRITTEN INTO THE OUTPUT FILE “PDSTRIP.OUT”.

8	32.060	8.300	0.800	0.600
9	32.118	8.300	0.800	0.600
10	32.128	8.300	0.800	0.600
11	32.120	8.300	0.800	0.600
12	32.112	8.300	0.800	0.600
13	32.080	8.300	0.800	0.600
14	32.034	8.300	0.800	0.600
15	31.978	8.300	0.800	0.600
16	31.822	8.300	0.800	0.600
17	31.186	8.300	0.800	0.600
18	29.430	8.300	0.800	0.600
19	26.230	8.300	0.800	0.600
20	21.700	8.300	0.800	0.600
21	16.294	8.300	0.800	0.600
22	10.614	8.300	0.800	0.600
23	5.736	8.300	0.800	0.600
24	5.504	8.300	0.800	0.600
25	5.628	8.300	0.800	0.600
26	5.968	8.300	0.800	0.600
27	6.102	4.990	0.800	0.600

Transverse metacentric height: 4.536

No fins

No sails

No forces depending on ship motions

No suspended weight

In the following 1,2,3 designate directions forward, to starboard, downward

Number of frequencies 99

<

*THE METACENTRIC HEIGHT DETERMINED BY PDSTRIP MAY DEVIATE FROM THE CHOSEN VALUE BECAUSE OF DISCRETIZATION ERRORS. THE TRANSVERSE METACENTRIC HEIGHT **GM** INDICATED IN THE OUTPUT LIST IS THEN TO BE CHECKED AND THE VERTICAL COORDINATE OF THE CENTRE OF GRAVITY **ZG** TO BE CHANGED, IF NECESSARY, SO THAT **GM** BECOMES CORRECT.*

RESULTS ARE WRITTEN INTO THE OUTPUT FILE “PDSTRIP.OUT”.

```
Yaw drift moment per wave amplitude squared -0.426E+05
Long., transv. reduced water drift velocity per wave amplitude^2 3134.163 -1809.508
-----
Wave circ. frequency 0.080 encounter frequ. 0.080 wave length 1000.00 wave number 0.0063 wave angle 0.0
speed 0.00 wetted transom? T log(determinant) 164.85 3.15
Real part(1) Imagin.part(1) Abs(1) Real part(2) Imagin.part(2) Abs(2) Real part(3) Imagin.part(3) Abs(3)
Translation 256.000 -8191.999 8195.998 1.072 -118.512 118.517 0.499 0.013 0.499
Rotation/k *****
0.116 0.526 0.539 3.845 136.881 136.935
-----
MassAj.
-----
9.0967639E+02 0.0000000E+00 0.0000000E+00 0.0000000E+00 -3.5146626E+03 0.0000000E+00
0.0000000E+00 4.7095176E+04 -1.6360832E-02 -9.1865203E+04 -2.8039212E+00 6.3852856E+05
0.0000000E+00 -1.0294529E-01 6.2713695E+04 -1.6079147E-01 9.6858362E+05 3.9857082E+00
0.0000000E+00 -9.1945703E+04 3.7585041E-01 2.0544611E+06 1.3851483E+01 7.1175880E+06
-3.5146626E+03 -3.9857082E+00 9.6858356E+05 -1.0261330E+01 1.5975517E+08 1.5957420E+02
0.0000000E+00 6.3852869E+05 2.8039212E+00 7.1096645E+06 5.0483067E+01 1.3075589E+08
-----
Amorti.
-----
0.0000000E+00 0.0000000E+00 0.0000000E+00 0.0000000E+00 0.0000000E+00 0.0000000E+00
0.0000000E+00 -1.6141178E+14 4.4521610E-03 -1.9369414E+15 -3.5901245E-02 -8.2126518E+15
0.0000000E+00 2.0483034E-03 -3.2975739E+14 3.1424377E-02 9.4421828E+15 -4.1898455E-02
0.0000000E+00 -1.9369414E+15 -3.3942123E-03 -2.3243296E+16 1.3265945E+00 -9.8551821E+16
0.0000000E+00 4.1898463E-02 9.4421828E+15 1.8211250E-01 -1.0120259E+18 -3.1730523E+00
0.0000000E+00 -8.2126518E+15 3.5901252E-02 -9.8551821E+16 -1.4143766E+01 -8.0321551E+17
-----
Efforts
-----
6.2951645E+01 -4.7946343E-01 -2.1569290E+12 1.8776697E+17 2.5802281E+14 -5.0804295E+01
-2.1094762E+01 3.5794649E-01 -1.2607883E+13 1.8623575E+16 3.1786912E+14 3.1888447E+01
*** SURFRIDING. Linearization inappropriate
Longitudinal and transverse drift force per wave amplitude squared *****
Yaw drift moment per wave amplitude squared -0.356E+26
Long., transv. reduced water drift velocity per wave amplitude^2 3619.019 0.000
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```

IT IS IMPORTANT TO VERIFY THAT FOR EACH WAVE FREQUENCY AND WAVE-ANGLE, THE RESULTS ARE CORRECT. IN GENERAL, PROBLEMS HAPPEN WITH THE CALCULATION OF DRIFT FORCES AND THE MESSAGE “SURFRIDING. LINEARIZATION INAPPROPRIATE” APPEARS AND/OR IT WAS NOT POSSIBLE TO OBTAIN A CORRECT VALUE FOR THE DRIFT FORCES. AS PREVIOUSLY MENTIONED, IN THIS CASE, THE LOWER LIMIT OF FREQUENCY RANGE HAS TO BE INCREASED A LITTLE AND/OR THE UPPER LIMIT OF FREQUENCY RANGE TO BE DECREASED, DEPENDING ON WHICH FREQUENCY THE PROBLEM APPEARS.